Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/31/2022
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2016	**End Date:**	09/30/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2022 per NSSC information (Ed., 9/22/21)		
Key Personnel Changes/Previous PI:	June 2019 report: No change to report.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX16AT09G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: The NASA Task Book reflects the NSSC project title, "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism." The project is also known as "Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism" (Ed., 7/18/23). ---------- NOTE: Continuation with same Principal Investigator (PI) Jamal Yagoobi, Ph.D. of "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism," grant NNX12AR32G. Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction		

Task Description:	Mechanism is an International Space Station (ISS) Experiment to fundamentally investigate the effects of gravity on: 1.) electrically generated two-phase flow (liquid/vapor) and 2.) electrically driven liquid film flow boiling. The proposed work is a collaborative effort between the investigators to develop and validate the flight experiment hardware. We propose this effort based upon the successes of our current project: NASA Grant/Cooperative Agreement: NNX12AR32G Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism. Although the original effort extends through February 2017, we have completed tasks and achieved the technical and programmatic milestones of this project. That is, we have passed all the requirements of the Science Concept Review and received the authorization to advance to the next stage of the experiment development. Therefore, the PI and Co-PI submit this proposal to develop the flight experiment hardware with an effective start date of October 2016. This proposal fundamentally investigates the complex liquid-vapor phase change phenomenon in presence of electrical field and absence of gravity. Thus, this study will provide valuable scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity. This is a proposal to support the development of the experiment flight hardware for the International Space Station, conduct the experiment on–board the International Space Station, support the operations of the experiment and evaluate the experimental data. The proposed period of performance is October 2016 to September 2021. However, this project was extended at no-cost till September 30, 2022. The robust collaboration between Worcester Polytechnic Institute (WPI) and Goddard Space Flight Center (GSFC) can provide both state of the art theoretical investigation and deliver flight qualified experimental hardware; thus, this proposal presents both investigative and hardware development tasks. The PI will maintain overall technical and scientific direction of the experiment development; however, the Co-I will support the flight experimental hardware development in collaboration with Glenn Research Center. Note: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status as of October 1, 2016, with this continuation grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This multi-physics proposal will provide basic scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity. The electrically driven liquid film flow boiling data aboard ISS will determine the critical heat flux, onset of boiling, absence of hysteresis, and bubble characteristics in the absence of gravity. There are a number of potential terrestrial and space applications for this electrohydrodynamic (EHD) driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices. To achieve these applications, the basic scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity is needed.
Task Progress:	Detailed experimental work was carried out in the Multi Scale Heat Transfer Laboratory at Worcester Polytechnic Institute (WPI) to understand the following: a- The effect of electrohydrodynamic (EHD) conduction pumping on liquid film flow boiling. b- The effect of dielectrophoretic (DEP) force on liquid film flow boiling. c- The combined effect of EHD conduction pumping and DEP force on liquid film flow boiling. The electric breakdown limit of HFE 7100 vapor was experimentally determined using the International Space Station (ISS) science unit at WPI’s multi scale heat transfer lab. The purpose of this testing was to determine how much voltage could be applied to the EHD conduction pump electrodes before breaking down the HFE 7100 vapor present. Fundamental analytical and experimental work was conducted to understand how the temperature of the working fluid impacts the pumping performance of an EHD conduction pump. Finally, a numerical analysis of the electrostatic properties of Parylene was conducted, prompted by a switch from Aeroglaze to Parylene for masking the EHD and DEP electrodes. The analysis proved the viability of Parylene as a suitable material for this purpose.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Abstracts for Journals and Proceedings	O’Connor NJ, Yagoobi J. "The effect of temperature on the pumping performance of a helical EHD conduction pump." 21st IEEE International Conference on Dielectric Liquids, Seville, Spain, May 29-June 2, 2022. Abstracts. 21st IEEE International Conference on Dielectric Liquids, Seville, Spain, May 29-June 2, 2022. , May-2022
Abstracts for Journals and Proceedings	Castaneda AJ, O’Connor NJ, Yagoobi J. "Dielectrophoretically assisted EHD driven liquid film flow boiling." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Abstracts for Journals and Proceedings	O’Connor NJ, Castaneda AJ, Yagoobi J. " Fundamental research in electrically driven liquid film flow." International Symposium on Electrohydrodynamics (ISEHD 2022), Otaru, Japan, August 28-September 2, 2022. Abstracts. International Symposium on Electrohydrodynamics (ISEHD 2022), Otaru, Japan, August 28-September 2, 2022. , Aug-2022

Articles in Peer-reviewed Journals	Castaneda AJ, O'Connor NJ, Yagoobi J, Didion J, Martins M, Hasan M. "Dielectrophoretically-assisted EHD-driven liquid film flow boiling in the presence and absence of gravity." J. Heat Mass Transfer. 2023 Mar;145(3):031601. https://doi.org/10.1115/1.4055566 , Mar-2023
Papers from Meeting Proceedings	O'Connor NJ, Yagoobi J. "An innovative electrohydrodynamic conduction pumping design for swirl flow generation." 56th IEEE Industry Applications Society Annual Meeting, Virtual, October 10-14, 2021. 56th IEEE Industry Applications Society Annual Meeting, Virtual, October 10-14, 2021. Paper #4: S26-P4-EPC. https://doi.org/10.1109/IAS48185.2021.9677332 , Oct-2021